

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Comparative Analysis

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO4: Articulation (BL4, BL5)	Assessment:	Assessment Tool 3 – Comparative Analysis
Weightage:	10% of CIA	Total Marks:	100
CO4 Statement:	Solve optimization problems using dynamic programming and greedy strategies.		
Student Name:	N.Srikanth	Reg. No.:	192411139
Section / Batch:	AI&ML	Date:	22/08/2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Analyze the given questions thoroughly before answering.
- Compare multiple aspects such as methods, results, advantages, and limitations clearly.
- Critically evaluate the strengths, weaknesses, and implications with proper justification.
- Present meaningful insights, interpretations, and well-supported conclusions from the comparisons.

Question Type	Focus Area	Questions
Comparative Analysis	Comparative analysis of Dynamic Programming and Greedy algorithms for optimization problems.	Q1

SECTION A – Comparative Analysis

Code of Conduct

I N.Srikanth(192411139) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

Aim

To find the minimum number of scalar multiplications required to multiply a chain of matrices using the Dynamic Programming approach.

Algorithm

1. Start the program.
2. Define the matrix dimensions in an array.
3. Initialize the Dynamic Programming table dp with 0.
4. Calculate the minimum cost for matrix chains of different lengths.
5. For each matrix chain, try all possible positions to divide the chain.
6. Calculate the multiplication cost and store the minimum cost in the dp table.
7. Display the minimum multiplication cost.
8. Stop the program.

Code

```
#include <iostream>
#include <climits>
using namespace std;

int main() {
    int p[] = {10, 20, 30, 40};
    int n = 3;
    int dp[4][4] = {};

    for (int l = 2; l <= n; l++) {
        for (int i = 1; i <= n - l + 1; i++) {
            int j = i + l - 1;
            dp[i][j] = INT_MAX;

            for (int k = i; k < j; k++) {
                dp[i][j] = min(dp[i][j],
                    dp[i][k] + dp[k + 1][j] +
                    p[i - 1] * p[k] * p[j]);
            }
        }
    }

    cout << "Minimum cost = " << dp[1][n];

    return 0;
}
```

Q1 Matrix Chain Multiplication (DP vs Greedy)

100 marks

Given matrices with dimensions (10×20), (20×30), (30×40), compare Dynamic Programming with a naive greedy approach for multiplication order. Analyze cost differences and correctness. Evaluate why greedy fails. Derive insights on optimization strategies.

Your Code

```
1 #include <iostream>
2 #include <climits>
3 using namespace std;
4
5 int main() {
6     int p[] = {10, 20, 30, 40};
7     int n = 3, dp[3][3] = {};
8
9     for (int l = 2; l <= n; l++)
10         for (int i = 1; i <= n-l+1; i++) {
11             int j = i+l-1;
12             dp[i][j] = INT_MAX;
13             for (int k = i; k < j; k++)
14                 dp[i][j] = min(dp[i][j],
15                                dp[i][k] + dp[k+1][j] + p[i-1]*p[k]*p[j]);
16         }
17
18     cout << "Minimum cost = " << dp[1][n];
19     return 0;
20 }
```

Input

```
p[] = {10, 20, 30, 40}
n = 3
```

Output

```
Minimum cost = 18001
```

RUBRICS FOR CALCULATION

Comparative Analysis					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Selection of Studies/Parameters	20%	Selects highly relevant and diverse studies with well-defined comparison parameters	Selects relevant studies with minor gaps in parameter definition	Limited or partially relevant selection	Poor or inappropriate selection of studies
Comparison & Differentiation	25%	Clearly compares multiple aspects (methods, results, limitations) with strong differentiation	Good comparison with minor gaps	Basic comparison with limited depth	No clear comparison; mostly descriptive
Critical Evaluation	25%	Critically evaluates strengths, weaknesses, and implications with strong justification	Provides evaluation with some justification	Limited evaluation; lacks depth	No critical evaluation provided
Synthesis & Insight Generation	20%	Generates meaningful insights and conclusions from comparisons	Some insights derived from comparison	Limited or unclear insights	No insights generated
Clarity	10%	Information is clearly presented (tables/figures) with logical structure	Generally clear with minor issues	Presentation lacks clarity or structure	Poor presentation and difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Selection of Studies/Parameters	20%				
Comparison & Differentiation	25%				
Critical Evaluation	25%				
Synthesis & Insight Generation	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:

Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____